

Weekly Self-Study Tasks to Complete

Phase 2 — Week 2: TypeScript, NPM, and Vite for Leaders (Part 2 of 2): Vite and Build Tooling

This document covers Part 2 of the two-week TypeScript, NPM, and Vite course, in the 4-hour Leaders model. Part 1 introduced TypeScript and NPM; this part shifts to the toolchain that turns typed source code into an instant developer experience and an optimized production build. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section is hands-on Vite exploration. Additional Content at the end is optional and sits outside the 4-hour budget. Every link and duration below has been verified against the source platform.

Core Resources (~60 min)

Watch the specific resources called out below — not entire courses. The goal is to understand why a build tool exists and what Vite does for your engineers, not to master its configuration. Keep a terminal open so the commands the video shows are real to you.

Topic	Material Description	Est. Time	Link / Reference
Modern Build Tools	Modern JavaScript Build Tools (Maaïke van Putten, Pluralsight, updated Jul 23, 2025; 1h 14m course). Watch “Understanding Modern JavaScript Build Tools” (~10 min), “Getting Started with Vite” (~8 min), and “Choosing a Build Tool” (~10 min). Why build tools exist, what Vite does differently, and how teams pick one — concept-first, leaders-appropriate.	~28 min	pluralsight.com — Modern JavaScript Build Tools
Why Vite	Vite “Why Vite” and “Features” (official docs) — read and skim how	~17 min	vite.dev — Why Vite

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	the dev server serves native ES modules for instant startup, plus Hot Module Replacement, TypeScript, and CSS handling. Free and authoritative.		
Build & Environment	Vite “Building for Production” and “Env Variables and Modes” (official docs) – read how npm run build produces an optimized dist/ folder and how VITE_ environment variables are exposed to the app. Free and authoritative.	~15 min	vite.dev – Building for Production

Practice Activities (1 hour)

Spend about an hour getting hands-on with Vite using the provided evergreen-quote-vite-tooling project on your lab VM. You will not be graded – the goal is to feel the workflow your engineers use on every modern front-end project.

Try the following, at your own pace:

- Open the provided evergreen-quote-vite-tooling project, run npm install, then npm run dev, and open the URL it prints (for example http://localhost:5173/) – the Evergreen page appears with a “Count a view” button.
- Click “Count a view” a few times, then edit the tagline in src/branding.ts and save – watch Hot Module Replacement update the page instantly, with no full reload and the view counter keeping its value.
- Open package.json and match each npm script – dev, build, and preview – to what it does.
- Run npm run build, look inside the generated dist/ folder (an index.html plus a hashed assets/ bundle for cache-busting), then run npm run preview to serve the production build.
- Create a .env file with VITE_APP_TITLE, then read it in src/App.tsx with import.meta.env.VITE_APP_TITLE – restart the dev server and watch the heading pick up the new title.

- Add a path alias (@ → src/) in both vite.config.ts and tsconfig.json, then switch the branding import in src/App.tsx to @/branding and confirm it still runs.
- Optional: in a different folder, scaffold a brand-new project with npm create vite@latest my-app -- --template react-ts, then npm install and npm run dev to see what Vite generates from scratch.

Budget about one hour and work at your own pace. By next week (React) you should be able to scaffold a Vite + React + TypeScript project from memory and explain what npm run build does.

Additional Content (Optional)

The following resources are optional and recommended for leaders who want to go deeper beyond the core curriculum. They sit outside the 4-hour weekly budget. React (taught with TypeScript) is covered next, so focus this week on the dev server, the production build, and project structure.

Topic	Material Description	Link / Reference
Vite + React	Using Vite with React (Zachary Bennett, Pluralsight, updated May 5, 2025; 35m) — a short, beginner-level course on setting up Vite with React, Hot Module Replacement, and production builds (Video Course).	pluralsight.com — Using Vite with React
Reference	Vite documentation — the authoritative guide to the dev server, configuration, plugins, and production builds (Reference).	vite.dev
Assets & CSS	Vite “Features” — how Vite handles CSS, CSS Modules, static assets, and the public/ directory (Reference).	vite.dev/guide/features
Strict TypeScript	The tsconfig “strict” reference — what strict mode turns on (noImplicitAny, strictNullChecks) and why teams enable it (Reference).	typescriptlang.org — tsconfig strict

Topic	Material Description	Link / Reference
Linting	typescript-eslint — the official ESLint tooling for TypeScript, for catching code-quality issues beyond type errors (Reference).	typescript-eslint.io